Memory

2025 Semester 1 SOFTENG 370: Operating Systems

Lecture 14

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1.0.0

Static Allocation is the Simplest Strategy

Create a fixed size allocation in your program
e.g. `char buffer[4096];`

When the program loads, the kernel sets aside that memory for you

That memory exists as long as your process does, no need to free

Dynamic Allocation is Often Required

- You may only conditionally require memory
  - Static allocations are sometimes wasteful
- You may not know the size of the allocation
  - Static allocations need to account for the maximum size
- Where do you allocate memory?
  - You can either allocate on the stack or on the heap

Variable sized allocation

Given a block of memory, how do we allocate it to satisfy various memory allocation requests?

This problem must be solved in the C library

- Allocates one or more pages from kernel via `brk`/`sbrk` or `mmap` system calls
- Gives out smaller chunks to user programs via malloc

This problem also occurs in the kernel

- Kernel must allocate memory for its internal data structures

Variable sized allocation: headers

Consider a simple implementation of malloc

Every allocated chunk has a header with info like size of chunk

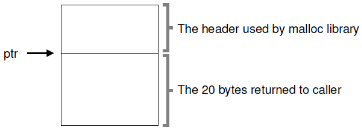


Figure 17.1: **An Allocated Region Plus Header**

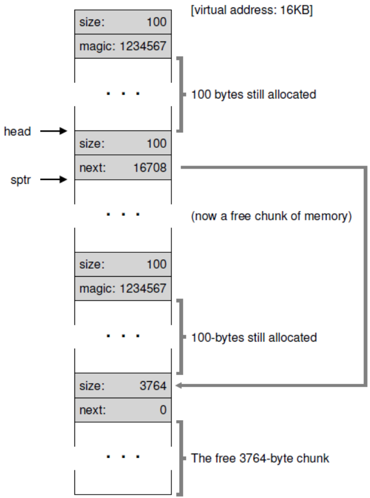
Free List

Free space is managed as a list

- Pointer to the next free chunk is embedded within the free chunk

The library tracks the head of the list

- Allocations happen from the head



Splitting and Coalescing

Suppose all the three chunks are freed

The list now has a bunch of free chunks that are adjacent

A smart algorithm would merge them all into a bigger free chunk

Must split and coalesce free chunks to satisfy variable sized requests

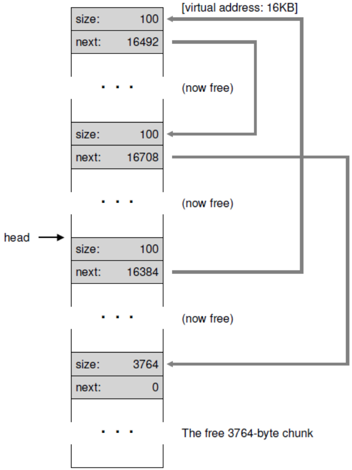


Figure 17.7: A Non-Coalesced Free List

Fragmentation is a Unique Issue for Dynamic Allocation

You allocate memory in different sized contiguous blocks

- Compaction is not possible and every allocation decision is permanent

A fragment is a small contiguous block of memory that cannot handle an allocation

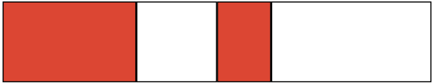
- You can think of it as a “hole” in memory, wasting space

There are 3 requirements for fragmentation

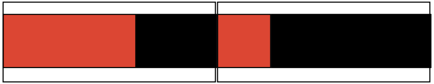
1. Different allocation lifetimes
2. Different allocation sizes
3. Inability to relocate previous allocations

There's Internal and External Fragmentation

External fragmentation occurs when you allocate different sized blocks
- There's no room for an allocation between the blocks



Internal fragmentation occurs when you allocate fixed sized blocks
- There's wasted space within a block



Credit: Daniel Ritz

We Want to Minimize Fragmentation

Fragmentation is just wasted space, which we should prevent

We want to reduce the number of “holes” between blocks of memory
- If we have holes, we’d like to keep them as large as possible

Our goal is to keep allocating memory without wasting space

Allocator Implementations Usually Use a Free List

- They keep track of free blocks of memory by chaining them together
  - Implemented with a linked list
- We need to be able to handle a request of any size
- For allocation, we choose a block large enough for the request
  - Remove it from the free list
- For deallocation, we add the block back to the free list
  - If it’s adjacent to another free block, we can merge them

There are 3 General Heap Allocation Strategies

- Best fit: choose the smallest block that can satisfy the request
  - Needs to search through the whole list (unless there’s an exact match)
- Worst fit: choose largest block (most leftover space)
  - Also has to search through the list
- First fit: choose first block that can satisfy request

Allocating Using Best Fit (1)

Note that blocks with a blank background and a number are free

Where do we allocate this block?

Allocating Using Best Fit (2)

Where do we allocate this block?

Allocating Using Best Fit (3)

The next block does not fit anywhere

Allocating Using Worst Fit (1)

Where do we allocate this block?

Allocating Using Worst Fit (2)

Where do we allocate this block?

Allocating Using Worst Fit (3)

Next block fits exactly in remaining space

Best Fit and Worst Fit are Both Slow

Best fit: tends to leave very large holes and very small holes
- Small holes may be useless

Worst fit: simulation says it's the worst in terms of storage utilization

First fit: tends to leave "average" size holes

The Buddy Allocator Restricts the Problem

- Typically, allocation requests are of size 2^n
  - e.g. 2, 4, 8, 16, 32, ..., 4096, ...
- Restrict allocations to be powers of 2 to enable a more efficient implementation
  - Split blocks into 2 until you can handle the request
- We want to be able to do fast searching and merging

You Can Implement the Buddy Allocator Using Multiple Lists

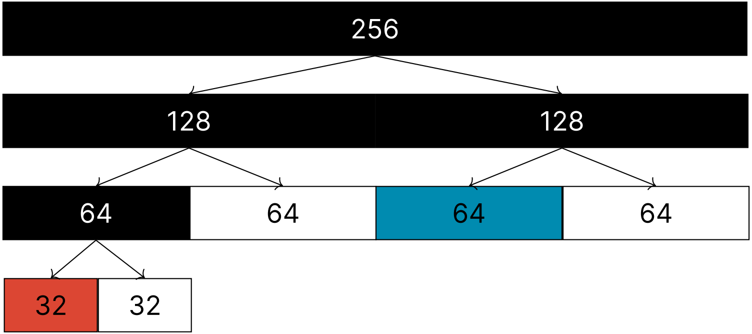
We restrict the requests to be $2^k, 0 \leq k \leq N$ (round up if needed)

Our implementation would use $N + 1$ free lists of blocks for each size

For a request of size 2^k, search the free list until we find a big enough block
- Search $k, k + 1, k + 2, ...$ until we find one
  - Recursively divide the block if needed until it's the correct size
  - Insert "buddy" blocks into free lists

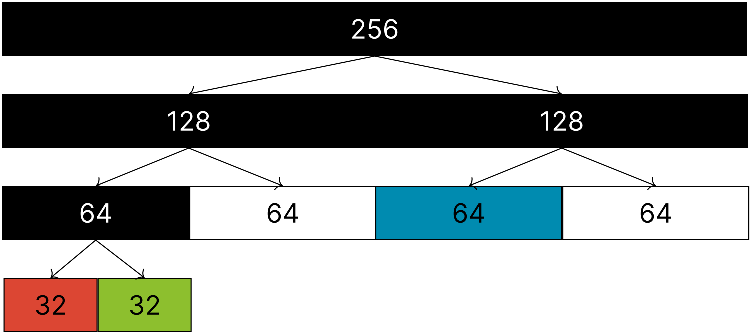
For deallocations, we coalesce the buddy blocks back together
- Recursively coalesce the blocks if needed

Using the Buddy Allocator (1)



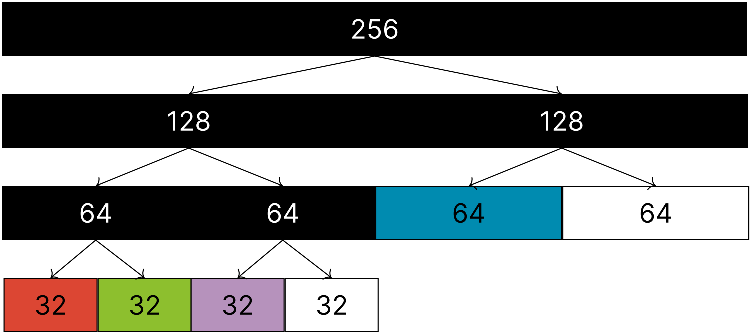
Where do we allocate a request of size 28?

Using the Buddy Allocator (2)



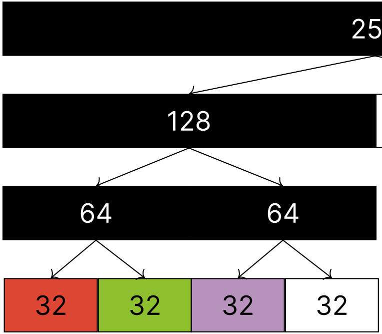
Where do we allocate a request of size 32?

Using the Buddy Allocator (3)



What happens when we free the size 64 block?

Using the Buddy Allocator (4)



Buddy Allocators are Used in Linux

- Advantages
  - Fast and simple compared to general dynamic memory allocation
  - Avoids external fragmentation by keeping free physical pages contiguous
- Disadvantages
  - There's always internal fragmentation
    - We always round up the allocation size if it's not a power of 2

Slab Allocators Take Advantage of Fixed Size Allocations

- Allocate objects of same size from a dedicated pool
  - All structures of the same type are the same size
- Every object type has it’s own pool with blocks of the correct size
  - This prevents internal fragmentation

Slab is a Cache of "Slots"

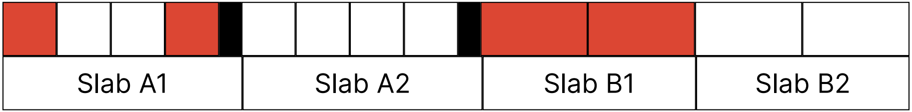
Each allocation size has a corresponding slab of slots
(one slot holds one allocation)

Instead of a linked list, we can use a bitmap
(there's a mapping between bit and slot)
- For allocations, we set the bit and return the slot
- For deallocations, we just clear the bit

The slab can be implemented on top of the buddy allocator

Each Slab Can Be Allocated using the Buddy Allocator

Consider two object sizes: A and B



We can reduce internal fragmentation if Slabs are located adjacently
- In this example A has internal fragmentation (dark box)